

Progressive block-size sweeps on QuickDraw

Five families $\times$ two inits $\times$ three rates ($m = n = 5, 32 \times 32$)

Mean test PSNR (dB) over 50 drawings. n/a = stage undefined (mera needs $k \in \{2, 4\}$ here).

1 A coarse-drawing regime

QuickDraw is 32×32 ($m = n = 5$), so the curriculum spans $k = 1..5$ (block sizes $2..32$; shown $k = 2..5$). The drawings are sparse strokes on a **dark** background. At this low resolution 8×8 -block transforms have little to exploit: classical block-DCT-8 reaches only 26.63 dB at $\rho = 0.20$, so the global learned circuits beat it. Setup otherwise matches the DIV2K / TU-Berlin matrix (1008 steps/stage, MSELoss top- k).

2 Light compression: $\rho = 0.20$ (5 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	35.2	35.5	35.1	35
rich	random	34.2	32.8	32.8	29.3
qft	identity	39.8	39.6	39.4	39.4
qft	random	33.4	32.5	30.1	30.1
tebd	identity	39.8	39.6	39.4	39.4
tebd	random	30	30	30	24.4
ent-qft	identity	39.8	39.6	39.4	39.4
ent-qft	random	30	30.1	30.1	24.4
mera	identity	39.8	n/a	39.4	n/a
mera	random	30	n/a	30	n/a

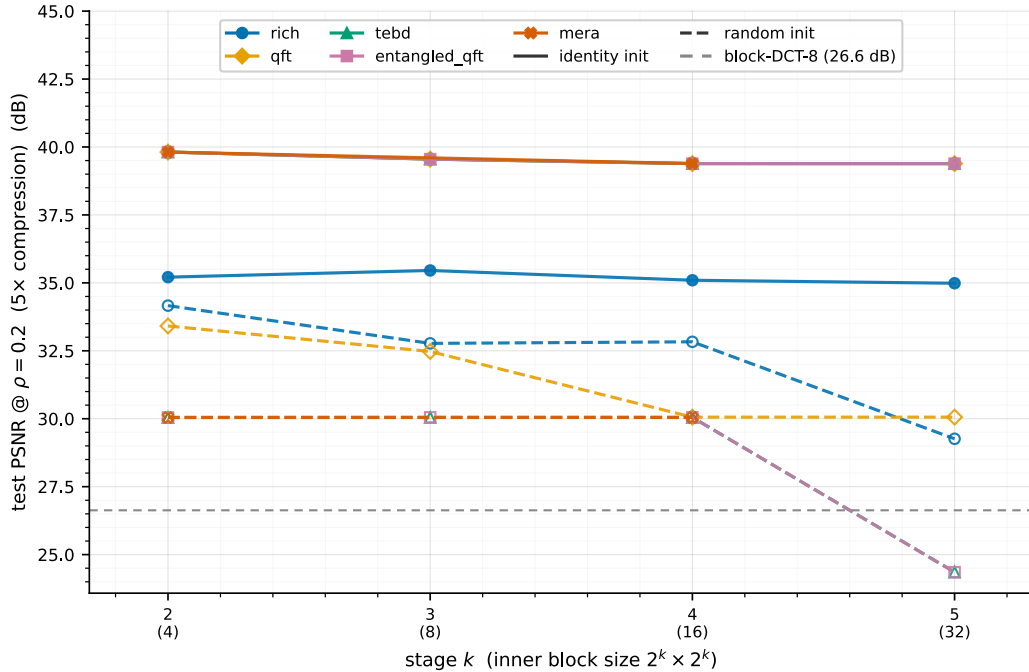


Figure 1: PSNR @ $\rho = 0.20$. The QFT-derived families (identity, all four bit-identical ≈ 39.5 dB) dominate — above rich (≈ 35) and well above block-DCT-8 (grey). Flat in k .

3 Heavy compression: $\rho = 0.05$ (20 $\times$)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	17.8	17.8	17.8	17.8

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	random	18.4	18.8	18.9	18.4
qft	identity	17.3	17.3	17.3	17.2
qft	random	17.8	17.3	18.1	18.1
tebd	identity	17.3	17.3	17.3	17.2
tebd	random	18.1	18.1	18.1	16.7
ent-qft	identity	17.3	17.3	17.3	17.2
ent-qft	random	18.1	18.1	18.1	16.7
mera	identity	17.3	n/a	17.3	n/a
mera	random	18.1	n/a	18.1	n/a

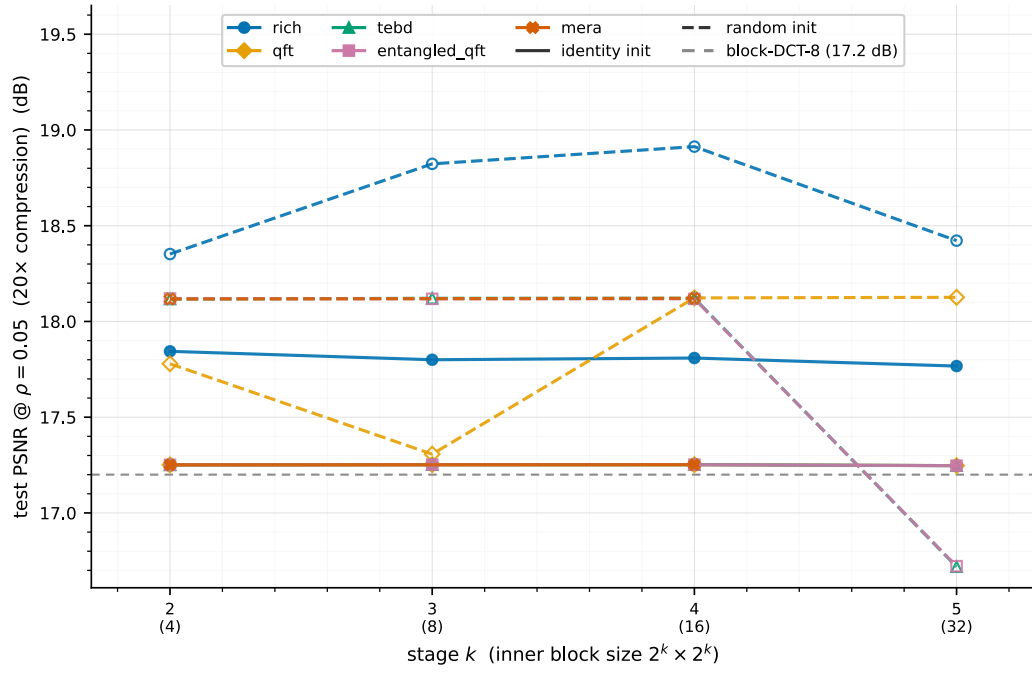


Figure 2: PSNR @ $\rho = 0.05$. All families bunch near block-DCT-8 (≈ 17 dB); the identity advantage vanishes, random init edges ahead at some k .

4 Very heavy compression: $\rho = 0.01$ (100x)

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	11.5	11.5	11.5	11.5
rich	random	12.2	12.7	12.8	13.3
qft	identity	11.3	11.3	11.3	11.3
qft	random	12.1	11.9	12.6	12.6
tebd	identity	11.3	11.3	11.3	11.3
tebd	random	12.6	12.6	12.6	13
ent-qft	identity	11.3	11.3	11.3	11.3
ent-qft	random	12.6	12.6	12.6	13
mera	identity	11.3	n/a	11.3	n/a
mera	random	12.6	n/a	12.6	n/a

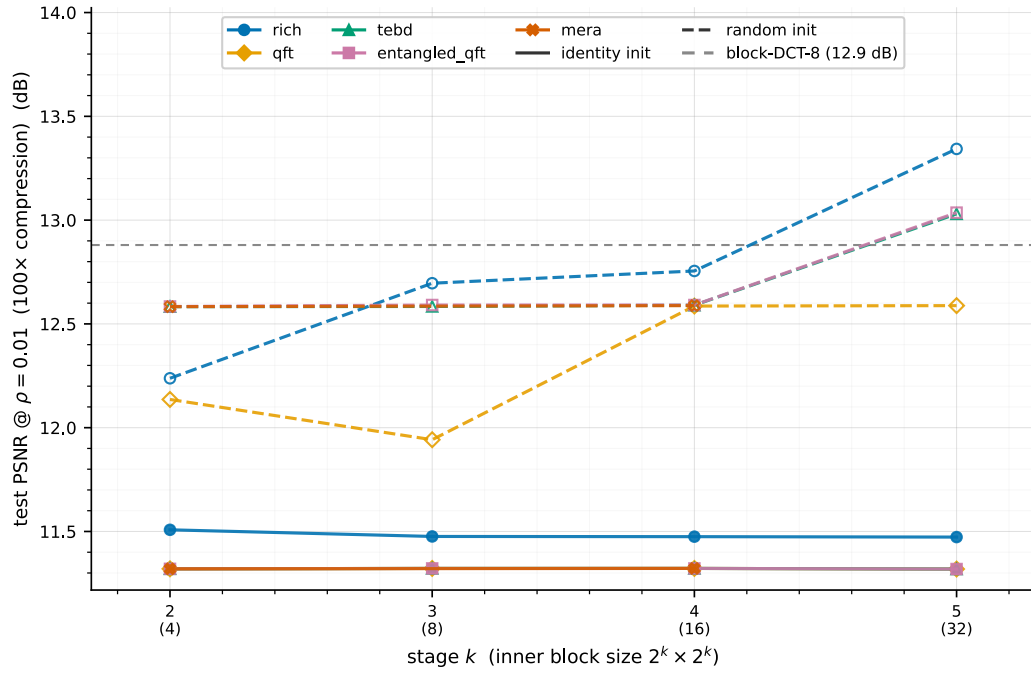


Figure 3: PSNR @ $\rho = 0.01$. Everything collapses toward block-DCT-8 (≈ 12.9 dB); families lie within ≈ 2 dB.

5 Observations

- **Learned circuits beat block-DCT at every rate (unlike the 256^2 sets)**, because 8×8 -block DCT is weak on 32×32 drawings — at $\rho = 0.20$ the QFT-derived families (≈ 39.5 dB) clear block-DCT (26.6) by ≈ 13 dB.
- **The QFT-family leads and is bit-identical under identity init**, beating rich by ≈ 4 dB at $\rho = 0.20$; the curve is flat in k (like DIV2K).
- **Identity $\triangleright$ random at the light rate only**; at $\rho = 0.05/0.01$ the families converge to block-DCT and the init advantage erodes (random sometimes edges ahead).

Working artefact. Combos from `results/family_init_matrix/quickdraw_5q/<family>_<init>/`; $\rho=0.01$ re-evaluated from checkpoints. Mean PSNR over the seed=42 / 50-drawing test split.